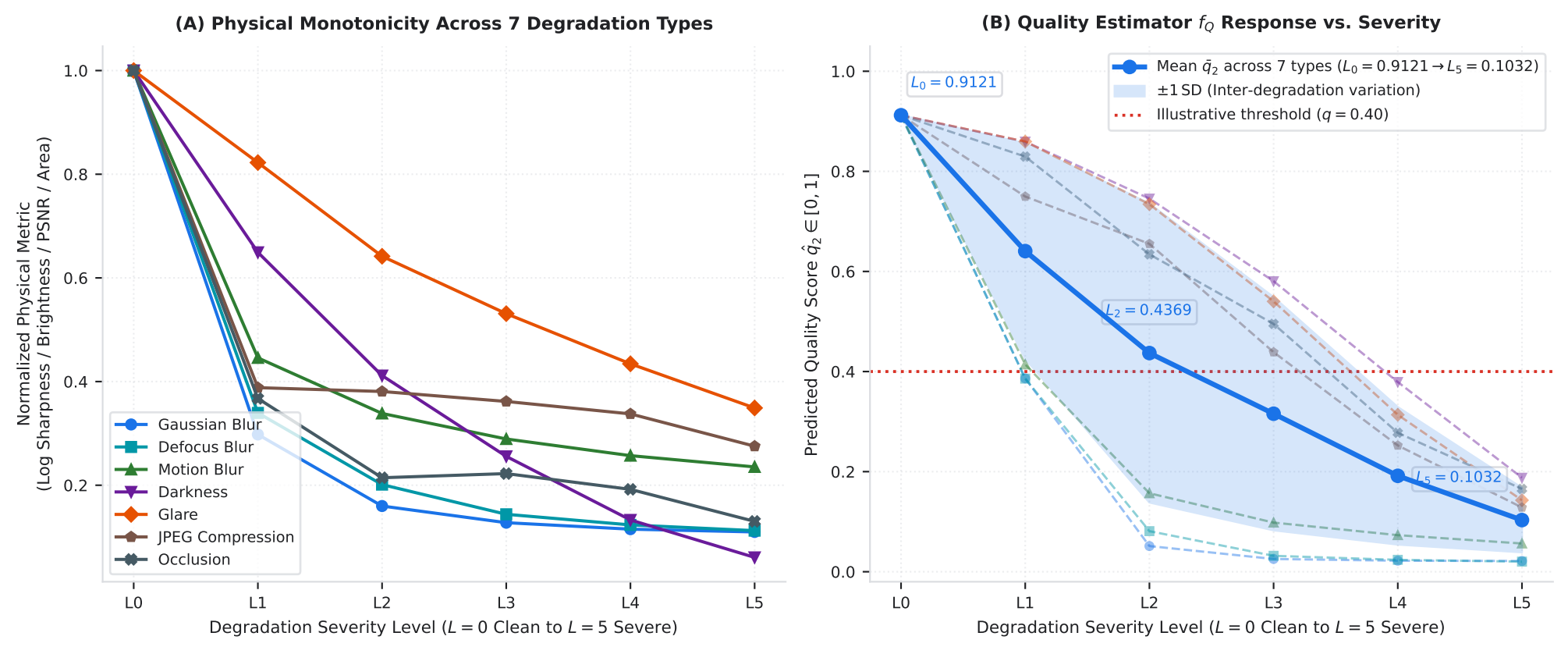
(A) Physical Monotonicity Across 7 Degradation Types
Normalized Physical Metric
(Log Sharpness / Brightness / PSNR / Area)
1.0
0.8
0.6
0.4
0.2
Gaussian Blur
Defocus Blur
Motion Blur
Darkness
Glare
JPEG Compression
Occlusion
L0
L1
L2
L3
L4
L5
Degradation Severity Level ($L = 0$ Clean to $L = 5$ Severe)
(B) Quality Estimator f_Q Response vs. Severity
Predicted Quality Score $\hat{q}_2 \in [0, 1]$
1.0
0.8
0.6
0.4
0.2
0.0
Mean $\bar{q}_2$ across 7 types ($L_0 = 0.9121 \rightarrow L_5 = 0.1032$)
± 1 SD (Inter-degradation variation)
Illustrative threshold ($q = 0.40$)
$L_0 = 0.9121$
$L_2 = 0.4369$
$L_5 = 0.1032$
L0
L1
L2
L3
L4
L5
Degradation Severity Level ($L = 0$ Clean to $L = 5$ Severe)